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SUPPLY VOLTAGE BASED ANALOG PREDISTORTION (APD) CIRCUIT FOR POWER AMPLIFIER

Abstract

A supply voltage based analog predistortion (APD) circuit for a power amplifier is disclosed. In an exemplary aspect, the power amplifier is in a front end module (FEM) of a radio frequency (RF) transceiver. An APD circuit operates within an amplifier chain to normalize the distortion profile of the amplifier chain based on a supply voltage. A baseband processor (BBP) performs digital predistortion (DPD) on signals being sent from the BBP to the FEM. As a result of the APD circuit, the DPD may assume a normalized profile for the FEM, allowing for simplification of the DPD despite many possible distortions introduced by the amplifier chain based on the supply voltage.

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Background/Summary

PRIORITY APPLICATION [0001] The present application is related to U.S. Provisional Patent Application Ser. 63/385,343 filed on Nov. 29, 2022, and entitled “SUPPLY VOLTAGE BASED ANALOG PREDISTORTION (APD) CIRCUIT FOR POWER AMPLIFIER,” the contents of which is incorporated herein by reference in its entirety. [0002] The present application is related to U.S. Provisional Patent Application Ser. 63/354,279 filed on Jun. 22, 2022, and entitled “FEMs WITH VCC DETECTION-BASED ANALOG-ASSISTED LINEARIZATION FOR DYNAMIC APT,” the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

I. Field of the Disclosure

[0003] The technology of the disclosure relates generally to providing linear operation for power amplifiers in a transmitter chain.

II. Background

[0004] Computing devices abound in modern society, and more particularly, mobile communication devices have become increasingly common. The prevalence of these mobile communication devices is driven in part by the many functions that are now enabled on such devices. Increased processing capabilities in such devices means that mobile communication devices have evolved from pure communication tools into sophisticated mobile entertainment centers, thus enabling enhanced user experiences. With the advent of the myriad functions available to such devices, there has been increased pressure to find ways to increase communication bandwidth available to supply data to the mobile communication device. This pressure has resulted in a trend to higher frequencies in the evolving cellular standards. These higher frequencies place additional pressure on the power amplifiers within the mobile communication devices to retain linear operation over a wide frequency range. Thus, there is opportunity for innovation in helping the power amplifiers improve linear operation, particularly at high frequencies.

SUMMARY

[0005] Aspects disclosed in the detailed description include a supply voltage based analog predistortion (APD) circuit for a power amplifier. In an exemplary aspect, the power amplifier is in a front end module (FEM) of a radio frequency (RF) transceiver. An APD circuit operates within an amplifier chain to normalize the distortion profile of the amplifier chain based on a supply voltage. A baseband processor (BBP) performs digital predistortion (DPD) on signals being sent from the BBP to the FEM. As a result of the APD circuit, the DPD may assume a normalized profile for the FEM, allowing for simplification of the DPD despite many possible distortions introduced by the amplifier chain based on the supply voltage.

[0006] In this regard in one aspect, a transmit chain is disclosed. The transmit chain comprises a power amplifier that imposes distortion on a signal to be transmitted as a function of supply voltage provided to the power amplifier. The transmit chain also comprises an APD circuit coupled to the power amplifier. The APD circuit is configured to normalize the distortion imposed on the signal by the power amplifier.

[0007] In another aspect, a method for providing predistortion to a power amplifier is disclosed. The method comprises receiving at a power amplifier a signal to be transmitted. The method also comprises providing APD to the power amplifier based on a supply voltage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a block diagram of a power amplifier within a transmitter chain that experiences variable distortion as a function of a supply voltage;

[0009] FIG. 2 is a graph of a range of distortions for power amplifiers for different supply voltages across a range of input power (P_{in}) highlighting the need for predistortion;

[0010] FIG. 3 is a graph of a tightened range of distortions for power amplifiers for different mobile terminals versus P_{in} after analog predistortion (APD) according to the system of FIG. 2;

[0011] FIG. 4 is a block diagram of an APD system for a power amplifier showing supply voltage detection and APD corrections;

[0012] FIG. 5 is a block diagram of an APD system similar to the APD system of FIG. 4, but with multiple look-up tables (LUTs) to assist in the predistortion;

[0013] FIG. 6 is a block diagram of an APD system similar to the APD system of FIG. 4, but with an equation-driven APD controller to avoid use of a LUT;

[0014] FIG. 7 is a block diagram showing a varactor being used to assist in applying APD in a transmitter chain in an exemplary aspect of the present disclosure;

[0015] FIG. 8 is a block diagram illustrating how APD may be spread across stages within a power amplifier and particularly spread across a hybrid complementary metal oxide semiconductor (CMOS) stage and a bipolar material portion amplifier;

[0016] FIG. 9 is a block diagram of an alternate exemplary aspect where all adjustments are calculated in a CMOS portion of a hybrid amplifier but made in both the CMOS portion and the bipolar material portion with two supply voltage detectors;

[0017] FIG. 10 is a block diagram of an alternate exemplary aspect, where gain distortion is addressed along with phase distortion;

[0018] FIG. 11 is a block diagram of a transmitter chain with additional details about how sampling of the supply voltage may be done to provide measurements for APD;

[0019] FIG. 12 illustrates two write signals with sampling triggers identified for a transmitter chain of FIG. 13;

[0020] FIG. 13 is a transmitter chain that uses the sampling triggers of FIG. 12;

[0021] FIG. 14 is a transmitter chain that uses additional information from a baseband processor (BBP) in evaluating the supply voltage when determining an APD to apply; and

[0022] FIG. 15 is a block diagram of a mobile terminal, which may include the predistortion systems of FIGS. 2-14 according to the present disclosure.

DETAILED DESCRIPTION

[0023] The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

[0024] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing

from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0025] It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0026] Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

[0027] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0028] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0029] Aspects disclosed in the detailed description include a supply voltage based analog predistortion (APD) circuit for a power amplifier. In an exemplary aspect, the power amplifier is in a front end module (FEM) of a radio frequency (RF) transceiver. An APD circuit operates within an amplifier chain to normalize the distortion profile of the amplifier chain based on a supply voltage. A baseband processor (BBP) performs digital predistortion (DPD) on signals being sent from the BBP to the FEM. As a result of the APD circuit, the DPD may assume a normalized profile for the FEM, allowing for simplification of the DPD despite many possible distortions introduced by the amplifier chain based on the supply voltage.

[0030] Before addressing exemplary aspects of the present disclosure, a brief overview of some challenges present in conventional devices are explored with reference to FIGS. 1 and 2. A discussion of exemplary aspects of the present disclosure begins below with reference to FIG. 3.

[0031] In this regard, FIG. 1 is an exemplary power amplifier stage **100** that may be present in a transmit chain. More specifically, the power amplifier stage **100** may have an input **102** that has a direct current (DC) blocking capacitor **104**. Signals arrive through the DC blocking capacitor **104** at an input of a power amplifier **106**. The power amplifier **106** may include bipolar transistors (shown) or complementary metal oxide semiconductor (CMOS) transistors (not shown). The power amplifier **106** may also receive a bias signal at the input through a bias resistor **108**. An output of

the power amplifier **106** may be coupled to an acoustic filter **110** and a supply voltage (VCC) **112**. The acoustic filter **110** may have a varying impedance that varies as a function of power levels. A capacitor **114** is effectively formed between a base and a collector of the transistor within the power amplifier **106**. The capacitance of this capacitor **114** may vary as a function of the supply voltage **112**. Such variation will induce a varying distortion in the output signal. Additional distortion may be induced by impedance fluctuations of the acoustic filter **110**. This distortion will primarily be in the phase of the output signal, but some distortion may occur in the amplitude of the output signal. [0032] The distortion will vary not only as a function of the supply voltage, but also as a function of the input power (i.e., the power presented at the input **102**) as better illustrated in FIG. 2, where graph **200** shows the phase distortion of three different supply voltages (VCC1-VCC3) in three curves **202(1)**-**202(3)**. Specifically, the y-axis **204** is the distortion and the x-axis **206** is the input power (Pin) at the input **102**.

[0033] In general, the distortion induced by the variation in the power has been addressed by providing digital predistortion (DPD) in a baseband processor (BBP). When the supply voltage is constant, a relatively small number of DPD coefficients may be needed. However, when the supply voltage is constant, it must be set high enough to accommodate any power peaks, which results in inefficiencies at lower power levels. This inefficiency led to the introduction of fast average power tracking (APT) circuits, which may change supply voltages in each slot and/or even for each symbol. However, each supply voltage will necessitate its own set of DPD coefficients. As the granularity of the supply voltages increases, the burden in terms of memory on the BBP increases. This burden translates to increased cost, increased space consumed, and potentially adds to latency.

[0034] Exemplary aspects of the present disclosure help alleviate the burden on the BBP by adding in APD circuitry to the power amplifier stage in a transmitter that normalizes the distortion caused by varying supply voltages. When the distortion profile caused by the varying supply voltages is normalized, the BBP may provide DPD that matches that normalized profile, obviating the need for numerous sets of DPD coefficients in the BBP as better seen in graph **300** of FIG. 3. In particular, graph **300** shows curves **202(1)** and **202(3)** being distorted by APD circuitry to be closer to curve **202(2)** allowing the BBP to have DPD coefficients for just the general shape of curve **202(2)**.

[0035] While the distortion is primarily phase distortion, both phase and gain distortion may be corrected with the APD circuitry of the present disclosure. Further, the APD circuitry of the present disclosure may be spread across multiple stages (e.g., a driver stage and an output stage) or across technologies (e.g., a bipolar transistor portion and CMOS transistor portion). The APD circuitry may adjust bias signals or otherwise modulate operation with varactors or the like. Likewise, knowledge of the supply voltage may be acquired through various sensors, signals from the BBP, and/or at certain timing intervals. All of these aspects are discussed in greater detail below.

[0036] In this regard, FIG. 4 illustrates a transmit chain **400** that includes a BBP **402** and an amplifier chain **404**. The BBP **402** includes a DPD circuit **406** with a memory **408** associated therewith that stores DPD coefficients. The memory **408** may be relatively small because a small set of DPD coefficients may be used by the BBP **402**. The BBP **402** generates a signal **410** to be transmitted as well as a control signal **412** for a power tracking circuit **414** in the amplifier chain **404**. In an exemplary aspect, the power tracking circuit **414** may be an APT circuit, and may be referred to as a fast APT, a dynamic APT, a symbol tracking circuit, or even a sub-symbol tracking circuit. The power tracking circuit **414** takes a known voltage level such as Vbat and converts this voltage level to a desired supply voltage (VCC) based on the control signal **412**. The supply voltage is provided through an impedance **416** to an output **418** of a power amplifier **420**. The output **418** is coupled to an antenna (not shown) through a filter **422**, which may be an acoustic filter.

[0037] The amplifier chain **404** may have multiple amplifier stages including the power amplifier **420** (which may be an output stage amplifier) and a driver stage amplifier **424**. A blocking capacitor **426** may separate the amplifiers **420**, **424**. Additional transmit circuitry (e.g.,

upconversion circuitry, switches, filters, or the like) **428** may also be present in the amplifier chain **404**. Still further, the amplifier chain **404** may include a digital input/output (I/O) **430** that receives additional signals or information from the BBP **402** such as mode information, channel information, or the like. The digital I/O **430** may also be associated with a control circuit **432**. Alternatively, the control circuit **432** may be incorporated into the digital I/O **430**.

[0038] As still another option, the control circuit **432** may be incorporated into an APD circuit **434**. The APD circuit **434** receives signals from one or more supply voltage detectors **436(1)-436(N)** and generates control signals that may modify the operation of one more of the amplifiers **420, 424** to apply distortion in such a manner that normalizes the overall distortion of the amplifier chain **404** to one of a few (e.g., one) predetermined distortion curves.

[0039] FIG. 5 shows a transmit chain **500** which highlights one way in which the APD circuit **434** may determine what control signals to generate. Specifically, the APD circuit **434** may have an associated memory **502** that stores a look-up table (LUT) **504** having multiple pages thereon corresponding to power levels and detected supply voltages. Based on these inputs, and optionally inputs from the control circuit **432**, appropriate entries in the LUT **504** are selected and used.

[0040] Alternatively, as illustrated in FIG. 6, a transmit chain **600** may have an APD circuit **434** that includes an equation-based control circuit **602** that calculates appropriate signals for each detected supply voltage. This approach may avoid a large LUT but may require additional computation circuitry.

[0041] While the transmit chain **400** relies on the supply voltage detectors **436(1)-436(N)** and transmit chains **500, 600** rely on a supply voltage detector **436**, the amplifier chain **404** of a transmit chain **700** in FIG. 7 may also receive supply voltage information from the BBP in a signal **702** passed through the digital I/O **430**. In some aspects, the information in the signal **702** may match the information in the control signal **412**. Further, FIG. 7 shows that the signals generated by the APD circuit **434** may adjust a varactor **704** that may operate with interstage circuitry **706** to provide desired APD. Note the varactor **704** may be a bank of parallel capacitors with selected ones switched in and out to provide a desired capacitance or other variable capacitor structure.

[0042] More detail about a possible implementation is provided in FIG. 8 where a transmit chain **800** is implemented across different technologies. Specifically, a front-end portion **802** of the amplifier chain **404** is implemented in CMOS and may operate in the digital domain. A back-end portion **804** may be implemented in a bipolar material (e.g., gallium arsenide (GaAs)) and may operate in the analog domain. The APD circuit **434** may use a digital-to-analog converter (DAC) **806** to turn a digital signal into an analog signal to control the varactor **704**. Likewise, a varactor **808** may be implemented in the back-end portion **804**. The varactor **808** may be controlled by a regulator **810**, which in turn receives an analog signal from a DAC **812** coupled to the APD circuit **434**. The varactor **808** may couple to a node **814** at an input of the power amplifier **420** (e.g., at a base of a bipolar transistor within the power amplifier **420**).

[0043] As an alternative, the supply voltage detectors **436(1)-436(N)** may be split across technologies as illustrated by transmit chain **900** in FIG. 9. More specifically, the supply voltage detector **436(1)** may be implemented in CMOS and provide a digital signal to the APD circuit **434**, which controls the varactor **704** as previously described. Additionally, the supply voltage detector **436(2)** may be implemented in bipolar technology and use an APD circuit **902** to control the varactor **808**.

[0044] The above discussion has not specified what types of distortion are corrected. It should be appreciated that the APD of the present disclosure may correct phase distortion (i.e., so called amplitude modulation (AM)-to-phase modulation (PM) (AM-PM) distortion) or gain distortion (i.e., so called AM-to-AM (AM-AM) distortion) or both. FIG. 10 illustrates a transmit chain **1000** that explicitly shows this latter possibility. Specifically, the APD circuit **434** may include an AM-PM circuit **1002** that controls the varactor **704** through the DAC **806**. The APD circuit **434** may further include an AM-AM circuit **1004** that controls a gain of the driver amplifier **424**. Likewise,

the APD circuit **902** may include an AM-PM circuit **1006** that controls the varactor **808** and an AM-AM circuit **1008** that controls the gain of the power amplifier **420**. While both APD circuits **434, 902** are shown with both AM-PM and AM-AM circuits, it should be appreciated that one APD circuit **434, 902** may correct phase and the other APD circuit correct gain or one APD circuit **434, 902** may correct both phase and gain while the other only corrects phase (or only corrects gain). Likewise, while shown as two separate APD circuits **434, 902**, it should be appreciated that in some aspects, a single APD circuit implemented in just CMOS or just bipolar technology may be present. Likewise, the APD circuits **434, 902** may use a LUT or equations as needed or desired.

[0045] The present disclosure has been silent so far on how the supply voltage detectors **436(1)-436(N)** work. While the supply voltage detectors **436(1)-436(N)** may be continuous, it is more likely that the supply voltage detectors **436(1)-436(N)** will sample periodically such as once per slot or once per symbol, depending, potentially, on how frequently the supply voltage is changed (i.e., how quickly the power tracking circuit **414** changes VCC may dictate how frequently VCC needs to be sampled. FIG. **11** shows one possibility for a transmit chain **1100** that has a sampling circuit **1102** that uses information from an edge detector **1104** and a delay circuit (not shown) to trigger the supply voltage detector **436**. While shown before the driver amplifier **424**, the edge detector **1104** may be positioned between the amplifiers **420, 424**.

[0046] FIG. **12** shows signals **1202** and **1204** which have per-slot and per-symbol trigger edges, respectively. The signals **1202, 1204** would be the signal generated by the power tracking circuit **414** (also sometimes referred to as a power management integrated circuit (PMIC)), and these edges may be what is detected by the supply voltage detector **436**.

[0047] When the signals **1202, 1204** are combined with the transmit chain **1100**, the result is shown in a transmit chain **1300** in FIG. **13**. Note that the trigger may also be provided directly from the BBP **402**.

[0048] Note that the information in the signal **702** may also be compared to information from the supply voltage detector **436** to see if the power tracking circuit **414** is undershooting or overshooting relative to the instruction from the BBP **402** as illustrated in FIG. **14** where a transmit chain **1400** includes a comparison circuit **1402** that compares a decoded target supply voltage to a measured supply voltage.

[0049] Note that various aspects of the present disclosure may be mixed and matched because many of the elements are not mutually exclusive. For example, the supply voltage detectors **436(1)-436(N)** may all be CMOS detectors or all bipolar detectors. Other variations not illustrated or specifically set forth still fall within the scope of the present disclosure.

[0050] With reference to FIG. **15**, the concepts described above may be implemented in various types of user elements **1500**, such as mobile terminals, smart watches, tablets, computers, navigation devices, access points, and like wireless communication devices that support wireless communications, such as cellular, wireless local area network (WLAN), Bluetooth, and near field communications. The user elements **1500** will generally include a control system **1502**, a BBP **1504**, transmit circuitry **1506** (which may include any of the transmit chains **400, 500, 600, 700, 800, 900, 1000, 1100, 1300, or 1400** described above), receive circuitry **1508**, antenna switching circuitry **1510**, multiple antennas **1512**, and user interface circuitry **1514**. In a non-limiting example, the control system **1502** can be a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC), as an example. In this regard, the control system **1502** can include at least a microprocessor(s), an embedded memory circuit(s), and a communication bus interface(s). The receive circuitry **1508** receives radio frequency (RF) signals via the antennas **1512** and through the antenna switching circuitry **1510** from one or more base stations. A low noise amplifier and a filter of the receive circuitry **1508** cooperate to amplify and remove broadband interference from the received signal for processing. Downconversion and digitization circuitry (not shown) will then downconvert the filtered, received signal to an intermediate or baseband frequency signal, which is then digitized into one or more digital streams

using analog-to-digital converter(s) (ADCs).

[0051] The BBP **1504** processes the digitized received signal to extract the information or data bits conveyed in the received signal. This processing typically comprises demodulation, decoding, and error correction operations, as will be discussed on greater detail below. The BBP **1504** is generally implemented in one or more digital signal processors (DSPs) and ASICs.

[0052] For transmission, the BBP **1504** receives digitized data, which may represent voice, data, or control information, from the control system **1502**, which it encodes for transmission. The encoded data is output to the transmit circuitry **1506**, where a DAC converts the digitally-encoded data into an analog signal and a modulator modulates the analog signal onto a carrier signal that is at a desired transmit frequency or frequencies. A power amplifier will amplify the modulated carrier signal to a level appropriate for transmission, and deliver the modulated carrier signal to the antennas **1512** through the antenna switching circuitry **1510**. The multiple antennas **1512** and the replicated transmit and receive circuitries **1506**, **1508** may provide spatial diversity. Modulation and processing details will be understood by those skilled in the art.

[0053] It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0054] The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus, the disclosure is not intended to be limited to the examples and designs described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A transmit chain comprising: a power amplifier that imposes amplifier distortion on a digitally predistorted signal to be transmitted, wherein the amplifier distortion is a function of supply voltage provided to the power amplifier; and an analog predistortion (APD) circuit coupled to the power amplifier, the APD circuit configured to normalize the amplifier distortion imposed on the digitally predistorted signal by the power amplifier such that digital predistortion (DPD) applied to create the digitally predistorted signal may be achieved with coefficients corresponding to normalized amplifier distortion.
2. The transmit chain of claim 1, wherein the APD circuit comprises a varactor coupled to the power amplifier.
3. The transmit chain of claim 1, further comprising a supply voltage detector coupled to an output of the power amplifier.
4. The transmit chain of claim 1, further comprising a second power amplifier coupled in series to the power amplifier.
5. The transmit chain of claim 4, wherein the APD circuit is further configured to predistort the second power amplifier to offset variations as a function of the supply voltage.

- 6.** The transmit chain of claim 1, further comprising a power tracking circuit configured to set the supply voltage responsive to a control signal from a baseband processor (BBP).
 - 7.** The transmit chain of claim 6, wherein the power tracking circuit is configured to set the supply voltage on a per-slot basis.
 - 8.** The transmit chain of claim 6, wherein the power tracking circuit is configured to set the supply voltage on a per-symbol basis.
 - 9.** The transmit chain of claim 3, wherein the supply voltage detector is configured to sample a voltage level on a per-slot basis.
 - 10.** The transmit chain of claim 3, wherein the supply voltage detector is configured to sample a voltage level on a per-symbol basis.
 - 11.** The transmit chain of claim 1, further comprising a baseband processor (BBP) coupled to the power amplifier and configured to provide the signal to be transmitted.
 - 12.** The transmit chain of claim 11, wherein the BBP comprises a digital predistortion (DPD) circuit configured to provide the digitally predistorted signal that offsets a normalized distortion profile of the power amplifier.
 - 13.** The transmit chain of claim 1, wherein the power amplifier comprises a bipolar transistor.
 - 14.** The transmit chain of claim 1, wherein the APD circuit is configured to provide phase distortion APD.
 - 15.** The transmit chain of claim 1, wherein the APD circuit is configured to provide gain distortion APD.
 - 16.** The transmit chain of claim 3, wherein the supply voltage detector is configured to detect voltage levels continuously.
 - 17.** A method for providing predistortion to a power amplifier, comprising: receiving at a power amplifier a digitally predistorted signal to be transmitted; and providing analog predistortion (APD) to the power amplifier based on a supply voltage, wherein the APD normalizes the digitally predistorted signal such that digital predistortion, DPD, applied to create the digitally predistorted signal may be achieved with coefficients corresponding to normalized amplifier distortion.
 - 18.** The method of claim 17, further comprising providing the DPD on the signal to be transmitted at a baseband processor (BBP) prior to receiving at the power amplifier.
 - 19.** The method of claim 17, wherein providing the APD comprises providing phase distortion to the signal to be transmitted.
 - 20.** The method of claim 17, wherein providing the APD comprises providing gain distortion to the signal to be transmitted.
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